CHAPTER 1

LIMIT AND CONTINUITY

Informal Definition of a Limit

The precise definition of limit is somewhat technical, so it's easier to start with an informal definition.

Definition: Let $f(x)$ is defined for all x near c, (but not equal to c itself) and let Lbe any real number. Then, we call that L is the limit of *f(x)* as *x* approaches c if *f(x)* becomes close to $L$and we write

$$\lim_{x\to c} f(x) = L \text{or } f(x) \to L \text{ as } x \to c$$

- Note that, the limit definition has nothing to do with the value of *f(x)* when $x = c$(which may or may not exist). That is, the definition is that the value of *f(x)* approaches the number L as *x* approaches the number c (on either side of *c*), but $x \neq c$.

Notation: $\lim_{x\to c} f(x) = L$ which is usually read *f*(*x*) approaches L as *x* approaches *c*.

Example. Let $f(x) = x^2 + 5$. determine the limit of $f(x)$ as x approaches to 2.

Solution: The value of $f(x)$ as *x* approaches to 2 listed in the following table.

x	1.9	1.99	1.999	2.001	2.01	2.1
f(x)	8.6	8.9601	8.9960	9.004	9.04	9.4

From the above tables, we can conclude that

$$\lim_{x\to 2}(x^2 + 5) = 9$$

NOTE: The limit of the function does not depend on the value of the function at the point for which the limit is computed.

Example: How does the function $f(x) = \frac{x^2-1}{x-1}$ behave near x = 1?

Solution: The given formula defines f for all real numbers x except at $x \neq 1$ (we cannot divide by zero). For any x $\neq$ 1, we can simplify the formula by factoring the numerator and Canceling common factors:

$$f(x) = \frac{(x+1)(x-1)}{x-1} = x + 1 \quad for \quad x \neq 1$$

The graph of *f* is thus the line with the point (1, 2) removed. This removed point is shown as a "hole" in Figure 1.1. Even though $f(1)$ is not defined, it is clear that we can make the value of

f(x) as close as we want to 2 by choosing x close enough to 1 and we say that f(x) approaches the limit 2 as x approaches 1 and we write,

$$\lim_{x\to 1} f(x) = \lim_{x\to 1} \frac{x^2-1}{x-1} = 2$$

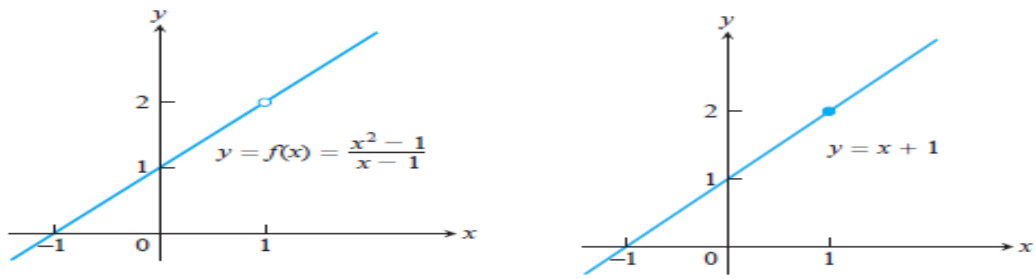


Figure 1.1

The graph of f is identical with the line $y = x + 1$ except at $x = 1$ where f is not defined.

<u>Formal Definition of Limits</u>

Let f be a function defined at each point of some open interval containing $\boldsymbol{a}$, possibly at $\boldsymbol{a}$ itself,then a number $\boldsymbol{L}$ is the limit of $f(x)$ as $\boldsymbol{x}$ approaches $\boldsymbol{a}$ (or is the limit of f at $\boldsymbol{a}$) if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$If\ 0 < |x-a| < \delta,\ \ then\ f(x) - L| < \varepsilon.$$

If L is the limit of $f(x)$ as x approaches $\boldsymbol{a}$ then we write $\lim_{x\to a} f(x) = L$

Let us look at the following graph and let us also assume that the limit does exist

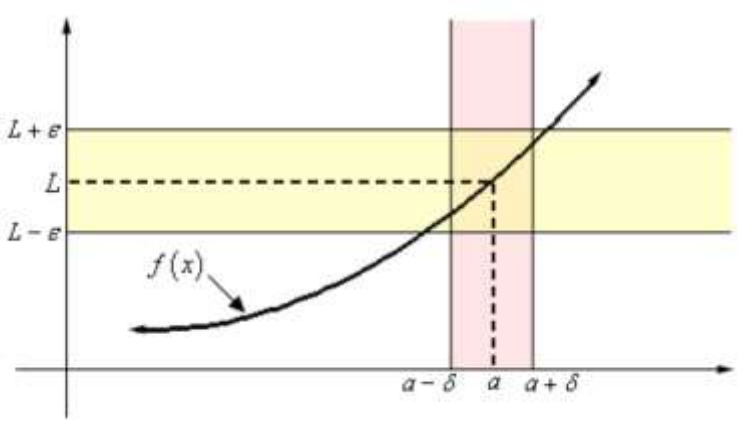


Figure 1.2

What the definition is telling us is for any number $\varepsilon > 0$ that we pick we can go to our graph and sketch two horizontal lines at $L + \varepsilon$ and $L - \varepsilon$ as shown on the graph above. Then somewhere out here in the world is another number $\delta > 0$. Which we will need to determine that will allow us to add in two vertical lines to our graph at $a + \delta$ and $a - \delta$.

Now, if we take any x in the pink region, i. e, between $a+\delta$ and $a-\delta$, then this x will be close to a than either of $a+\delta$ and $a-\delta$ or $|x-a|<\delta$.

If we now identify the point on the graph that our choice of x gives, then this point on the graph will lie in the intersection of the pink and yellow region. This means that this functional value of $f(x)$ will be close to L than either of $\mathrm{L}+\varepsilon$ and $\mathrm{L}-\varepsilon$, or $|f(x)-L|<\varepsilon$.

So, if we take value of x in the pink region then the graph for those values of x will lie in the yellow region.

Example: Use the definition of limit to prove the following limit

$$\lim_{x\to 3} 4x-7=5$$

Solution: We need to show that given $\varepsilon>0$ then there exists $\delta>0$ such that

$$\text{if} \quad 0<|x-3|<\delta \quad then \ |(4x-7)-5| < \varepsilon$$

To choose an appropriate δ we start with $|(4x-7)-5|<\varepsilon$ then we have

$$|4x-12|<\varepsilon \quad \Rightarrow \quad 4|x-3|<\varepsilon$$

Now we can Choice $\delta=\frac{\varepsilon}{4}$

Assume $0<|x-3|<\delta=\frac{\varepsilon}{4}$ and we have

$$|(4x-7)-5|=|4x-12| = 4|x-3|<4\delta=\frac{4\varepsilon}{4}=\varepsilon$$

$$|(4x-7)-5|<\varepsilon$$

Therefore, $\lim_{x\to 3} 4x-7=5$.

Example: Prove that $\lim_{x\to 3} x^2=9$

Solution: Let $\varepsilon>0$ be given, we have to find a number a $\delta>0$, such that If $0<|x-3|<\delta$ then $|x^2-9|<\varepsilon$

To connect $|x^2-9|$ with $|x-3|$ we write $|x^2-9|=|(x-3)(x+3)|$. Then we want

$$\text{If } 0<|x-3|<\delta \text{ then } |x+3|\,|x-3|<\varepsilon.$$

Notice that if we can find a positive constant C such that $|x+3|<C$ then

$$|x+3||x-3|<C|x-3|$$

And we can make $C|x-3|<\varepsilon$ by taking $|x-3|<\frac{\varepsilon}{C}=\delta$.

We can find such a number C if we restrict x to lie in some interval centered at 3.In fact . since we are interested only in values of x that are close to 3,it reasonable to assume that x is within a distance I from 3, that is , $|x-3|<1$. then $2<x<4$, so

$5 < x + 3 < 7$. Thus we have $|x + 3| < 7$, and so $C = 7$ is a suitable choice for the constant .

But now there are two restrictions on $|x + 3|$, namely

$$|x - 3| < 1 \text{ and } |x - 3| < \frac{\varepsilon}{C} = \frac{\varepsilon}{7}$$

To make sure that both of these inequalities are satisfied , we take δ to be the small of the two numbers 1 and $\frac{\varepsilon}{7}$. The notation for this is $\delta = \min\left\{1, \frac{\varepsilon}{7}\right\}$. This shows that $\lim_{x\to 3} x^2 = 9$.

Exercises

1. Use $\varepsilon - \delta$ definition proves the following limits.

a) $\lim_{x\to 2} 2x + 3 = 7$

b) $\lim_{x\to 10} \sqrt{x - 1} = 3$

c) $\lim_{x\to a} \sqrt{x} = \sqrt{a}$ *If* $a > 0$

$\left[Hint: Use \left|\sqrt{x} - \sqrt{a}\right| = \frac{|x-a|}{\sqrt{x}+\sqrt{a}}.\right]$

d). $\lim_{x\to 2} \frac{1}{x} = \frac{1}{2}$

e) $\lim_{x\to 4} x^2 = 16$

f) $\lim_{x\to k} k = k$

g) $\lim_{x\to 4} x^2 + x - 11 = 9$

h) $\lim_{x\to 4}(x - 4) = 0$

2. Find the number δ such that

a) If $|x - 2| < \delta$ *, then* $|4x - 8| < \varepsilon$, *where* $\varepsilon = 0.1$

b) Repeat part (a) with $\varepsilon = 0.01$

<u>Basic Limit Theorems</u>

Theorem:

i. If the limit of a function f at a exists, then this limit is unique.

ii. If L and M are both limits of f at a then $L = M$.

Theorem: Assume that $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exists and c is any constant.

i.e. $\lim_{x\to a} f(x) = K$ and $\lim_{x\to a} g(x) = L$ Then

1. $\lim_{x\to a}[f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = K + L$
2. $\lim_{x\to a}[f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x) = K - L$
3. $\lim_{x\to a} cf(x) = c \lim_{x\to a} f(X) = cK.$
4. $\lim_{x\to a} f(x)g(x) = \lim_{x\to a} f(x) \lim_{x\to a} g(x) = KL.$

5. If $\lim_{x\to a} g(x) \neq 0$ *and* $\lim_{x\to a} \frac{f(x)}{g(x)}$ *exist then*

$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} = \frac{K}{L}$.

6. $\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n = K^n$, where n is a positive integer.
7. $\lim_{x\to a} c = c$ and $\lim_{x\to a} x = a$
8. $\lim_{x\to a} x^n = a^n$, where n is a positive integer
9. $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$, where n is positive integer (if n is even, we assume that $a > 0$)
10. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$, where n is positive integer (if n is even. We assume that $a > 0$)
11. $\lim_{x\to a} e^{f(x)} = e^{\lim_{x\to a} f(x)} = e^L$
12. $\lim_{x\to a} (f(x))^{g(x)} = e^{\lim_{x\to a} g(x) ln|f(x)|}$ for $f(x) > 0$

Example. Suppose $\lim_{x\to a} f(x) = 4$ and $\lim_{x\to a} g(x) = -2$, then find;

a) $\lim_{x\to a}(f(x) + g(x))$ b) $\lim_{x\to a}(f(x)g(x))$ c) $\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right)$ d) $\lim_{x\to a} 4(g(x))$

2. Evaluate the following limits.

a) $\lim_{x\to 2} e^{\frac{x^2-4}{x-2}}$ b) $\lim_{x\to 1} sin^{-1}\left(\frac{1-\sqrt{x}}{1-x}\right)$ c) $\lim_{x\to \pi/2}(2 + \cos x)^{\sin x}$

Solution. a) $\lim_{x\to 2} e^{\frac{x^2-4}{x-2}} = e^{\lim_{x\to 2} \frac{x^2-4}{x-2}} = e^{\lim_{x\to 2} x+2} = e^4$.

b) $\lim_{x\to 1} sin^{-1}\left(\frac{1-\sqrt{x}}{1-x}\right) = sin^{-1}\left(\lim_{x\to 1} \frac{1-\sqrt{x}}{1-x}\right) = sin^{-1}\left(\lim_{x\to 1} \frac{1-\sqrt{x}}{(1-\sqrt{x})(1+\sqrt{x})}\right) = sin^{-1}\left(\lim_{x\to 1} \frac{1}{1+\sqrt{x}}\right) =$

$sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6} = 30°$.

c) $\lim_{x\to \pi/2}(2 + \cos x)^{\sin x} = e^{\lim_{x\to \pi/2} \sin x\, ln|2+\cos x|} = e^{\lim_{x\to \pi/2} \sin x \lim_{x\to \pi/2} ln|2+\cos x|} = e^{\ln 2} = 2$.

Theorem (squeezing theorem)

Assuming that $f(x) \leq g(x) \leq h(x)$ for all x in the some open interval I about a except possible a itself. If $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$, Then $\lim_{x\to a} g(x)$ exists $\lim_{x\to a} g(x) = L$

Exercises

1. Given that, $\lim_{x\to a} f(x) = -3$ $\lim_{x\to a} g(x) = 0$ $\lim_{x\to a} h(x) = 8$

Find the limits that exist if the limit does not exist, explain why.

a) $\lim_{x\to a}[f(x) + g(x)]$

b) $\lim_{x\to a}(f(x))^2$

c) $\lim_{x\to a}\sqrt{h(x)}$

d) $\lim_{x\to a}\frac{2f(x)}{h(x)-f(x)}$

e) $\lim_{x\to a}\sqrt{f(x)}$

f) $\lim_{x\to a}\frac{f(x)}{g(x)}$

2. Evaluate the limit and justify each step by indicating the appropriate limit laws.

a) $\lim_{x\to -2}(2x^4 + 5x^2 - 2x + 9)$

b) $\lim_{x\to 2}\frac{2x^2+1}{x^2-3x+4}$

c) $\lim_{t\to -1}(t^2+1)^3(t+3)^5$

d) $\lim_{x\to 8}\left(1+\sqrt[3]{x}\right)(2-3x^2+x^3)$

e) $\lim_{x\to 3}\frac{\sqrt{x^2+1}}{x+4} + \sqrt{16-x^2}$

f) $\lim_{x\to 2}(1+x^2)(2x-1)$

3. Evaluate the limit if it exists.

a) $\lim_{x\to 2}\frac{x^2+x-6}{x-2}$

b) $\lim_{x\to 1}\frac{x^3-1}{x^2-1}$

c) $\lim_{h\to 0}\frac{(4+h)^2-16}{h}$

d) $\lim_{h\to 0}\frac{\sqrt{1+h}-1}{h}$

e) $\lim_{t\to -3}\frac{t^2-9}{2t^{2+7t+3}}$

f) $\lim_{u\to -2}\sqrt{u^4+3u+6}$

g) $\lim_{x\to 1}\left(\frac{1+3x}{1+4x^2+3x^4}\right)^3$

h) $\lim_{t\to -1}(t^2+1)^3(t+1)^5$

4. *Use the squeezing theorem*

a) If $\sqrt{5-2x^2} \le f(x) \le \sqrt{5-2x^2}$ *for* $-1 \le x \le 1$ *find* $\lim_{x\to 0} f(x)$

b) If $2 - x^2 \le g(x) \le 2cosx$ *for all x, find* $\lim_{x\to 0} g(x)$

<u>One sided (left and right) limits</u>

- If $f(x)$ is defined on some interval (a, b)extending to the left of $x = b$ or right of $x = a$, and if we can ensure that $f(x)$ is as close as we want to L by taking x to the right of $a$or the left of $b$and close enough to a or b, then we say $f(x)$has right limit L or left limit L at $x = a$ or $x = b$ respectively, and we write

$$\lim_{x\to a^+} f(x) = L \text{ or } \lim_{x\to b^-} f(x) = L$$

- A function $f(x)$ has limit L at $x = a$ if and only if it has both left and right limits there and these one-sided limits are both equal to L:

$$\lim_{x\to a} f(x) = L \Leftrightarrow \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L$$

Example: Evaluate the following limits

1. If $f(x) = \frac{|x-2|}{x^2+x-6}$, find; $\lim_{x\to 2^+} f(x)$, $\lim_{x\to 2^-} f(x)$, and $\lim_{x\to 2} f(x)$.

Solution 1. Observe that $|x-2| = x-2$ if $x > 2$, and $|x-2| = -(x-2)$ if $x < 2$. Therefore,

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \frac{x-2}{x^2+x-6} = \lim_{x\to 2^+} \frac{x-2}{(x-2)(x+3)} = \lim_{x\to 2^+} \frac{1}{x+3} = \frac{1}{5} \text{and}$$

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} \frac{-(x-2)}{x^2+x-6} = \lim_{x\to 2^-} \frac{-(x-2)}{(x-2)(x+3)} = \lim_{x\to 2^-} \frac{-1}{x+3} = -\frac{1}{5}.$$

Since $\lim_{x\to 2^+} f(x) \neq \lim_{x\to 2^-} f(x)$**,** the limit $\lim_{x\to 2} f(x)$ does not exist.

2. Estimate the value of the limit $\lim_{x\to 2} g(x)$ where $g(x) = \begin{cases} \frac{x^2+4x-12}{x^2-2x} & if\ x \neq 2 \\ 6 & if\ x = 2 \end{cases}$

Solution: we've given a value for $x = 2$,so let's first note that $g(2) = 6$

To find the limit as x approaches **2** not exactly at **2** you must simplify the given functions

By using factorization and taking common factor where $g(x) = \frac{x^2+4x-12}{x^2-2x}$

Now calculate the limit $\lim_{x\to 2} g(x) = \lim_{x\to 2} \frac{(x+6)(x-2)}{x(x-2)}$

$= \lim_{x\to 2} \frac{x+6}{x}$ cancellation law

$= 4$

Exercises

1. *Find the limit, if it exists. If the limit does not exist, explain why.*

   a) $\lim_{x\to 3}(2x + |x-3|)$

   b) $\lim_{x\to -6} \frac{2x+12}{|x+6|}$

   c) $\lim_{x\to 0^-} \left(\frac{1}{x} - \frac{1}{|x|}\right)$

   d) $\lim_{x\to -2} \frac{2-|x|}{2+x}$

   e) $\lim_{x\to o^+} \left(\frac{1}{x} - \frac{1}{|x|}\right)$

   f) $\lim_{x\to 0.5^-} \frac{2x-1}{|3x^3-x^2|}$

   g) $\lim_{x\to 0^-} \frac{x}{|x|}$ *and* $\lim_{x\to 0^+} \frac{x}{|x|}$

   h) $\lim_{x\to 0} \frac{x}{|x|}$

2. *Let* $f(x) = \begin{cases} 4 - x^2 & if\ x \leq 2 \\ x - 1 & if\ \ x > 2 \end{cases}$ *Does* $\lim_{x\to 2} f(x)$ *exist?*

3. *Let* $f(x) = \frac{x^2-1}{|x-1|}$. *then find* $\lim_{x\to 1^+} f(x)$ *and* $\lim_{x\to 1^+} f(x)$

4. *Let* $f(x) = \begin{cases} -x & if\ x \le -1 \\ 1 - x^2 & if -1 < x < 1 \\ x - 1 & if x > 1 \end{cases}$ *Then evaluate the limit ,if it exists.*

a) $\lim_{x\to 1^+} f(x)$ *b)* $\lim_{x\to 1} f(x)$ *c)* $\lim_{x\to 0} f(x)$

d) $\lim_{x\to -1^-} f(x)$ *e)* $\lim_{x\to -1^+} f(x)$ *f)* $\lim_{x\to -1} f(x)$

Limits at Infinity and Infinite Limits

Limits at infinity: - Suppose that the function $\boldsymbol{f}$ is defined on an interval $(\boldsymbol{a}, \infty)$ and if we can ensure that $\boldsymbol{f}(\boldsymbol{x})$ is as close as we want to the number $\boldsymbol{L}$ by taking $\boldsymbol{x}$ large enough, then we say that $f(x)$approaches the limit L as x approaches infinity, and we write

$$\lim_{x\to\infty} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{L}$$

If f is defined on an interval $(-\infty, b)$ and if we can ensure that $f(x)$is as close as we want to the number M by taking x negative and large enough in absolute value, then we say that $f(x)$approaches the limit M as x approaches negative infinity, and we write

$$\lim_{x\to-\infty} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{M}$$

Example. Evaluate $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ for

$a.\ f(x) = \frac{x}{\sqrt{x^2+1}}$. b. $f(x) = \sqrt{x^2 + x} - x$ c. $f(x) = \sqrt{x} - \sqrt{x + \sqrt{x}}$

Solution 1.Rewrite the expression for $f(x)$ as follows:

$$f(x) = \frac{x}{\sqrt{x^2+1}} = \frac{x}{\sqrt{x^2(1+\frac{1}{x^2})}} = \frac{x}{\sqrt{x^2}\sqrt{1+\frac{1}{x^2}}} = \frac{x}{|x|\sqrt{1+\frac{1}{x^2}}}.$$

Now, we have that $\frac{x}{|x|} = \begin{cases} 1, & for\ x > 0 \\ -1 & for\ x < 0 \end{cases}$.

Therefore, $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x}{|x|\sqrt{1+\frac{1}{x^2}}} = \lim_{x\to\infty} \frac{1}{\sqrt{1+\frac{1}{x^2}}} = 1$ and similarly $\lim_{x\to-\infty} f(x) = -1$.

b. We are trying to find the limit of the difference of two functions, each of which becomes arbitrarily large as x increases to infinity. We rationalize the expression by multiplying the numerator and the denominator (which is 1) by the conjugate expression $\sqrt{x^2 + x} - x$:

$$\lim_{x\to\infty}\left(\sqrt{x^2 + x} - x\right) = \lim_{x\to\infty} \frac{\left(\sqrt{x^2 + x} - x\right)\left(\sqrt{x^2 + x} + x\right)}{\sqrt{x^2 + x} + x} = \lim_{x\to\infty} \frac{x^2 + x - x^2}{\sqrt{x^2(1 + \frac{1}{x})} + x}$$

$$= \lim_{x\to\infty} \frac{x}{x\sqrt{1+\frac{1}{x}}+x} = \lim_{x\to\infty} \frac{1}{\sqrt{1+\frac{1}{x}}+1} = \frac{1}{2}.$$

Limits at Infinity for Rational Functions

- The only polynomials that have limits at $\pm\infty$ are constant ones, $P(x) = c$. Recall that a rational function is a quotient of two polynomials.
- The way to do limits at positive infinity and negative infinity (if they exist) is to divide the numerator and denominator by the highest power of x appearing in the denominator.
- The limits of a rational function at positive infinity and negative infinity either both fail to exist or both exist and are equal.

Example. Evaluate the following limits a. $\lim_{x\to\pm\infty} \frac{2x^4-3x^2+1}{6x^4+x^3-3x}$ b. $f(x) = \frac{\sqrt{4x^2+2}}{3x+1}$ c . $\lim_{x\to\pm\infty} \frac{5x+2}{2x^3-1}$

Solution a: Divide the numerator and the denominator by x^4, the highest power of x appearing in the denominator:

$$\lim_{x\to\infty} \frac{2x^4-3x^2+1}{6x^4+x^3-3x} = \lim_{x\to\infty} \frac{2-3/x^2+1/x^4}{6+1/x-3/x^3} = \frac{\lim_{x\to\infty} 2 - \lim_{x\to\infty} 3/x^2 + \lim_{x\to\infty} 1/x^4}{\lim_{x\to\infty} 6 + \lim_{x\to\infty} 1/x - \lim_{x\to\infty} 3/x^3} = \frac{2}{6} = \frac{1}{3}.$$

b. $\lim_{x\to\infty} \frac{\sqrt{4x^2+2}}{3x+1} = \lim_{x\to\infty} \frac{\sqrt{x^2(4+\frac{2}{x^2})}}{x(3+\frac{1}{x})} = \lim_{x\to\infty} \frac{\sqrt{4+\frac{2}{x^2}}}{3+\frac{1}{x}} = \frac{\sqrt{\lim_{x\to\infty}(4+\frac{2}{x^2})}}{\lim_{x\to\infty}(3+\frac{1}{x})} = \frac{\sqrt{\lim_{x\to\infty} 4 + \lim_{x\to\infty}\frac{2}{x^2})}}{\lim_{x\to\infty} 3 + \lim_{x\to\infty}\frac{1}{x})} = \frac{\sqrt{4+0}}{3+0)} = \frac{2}{3}.$

Infinite Limits at Infinity

The notation $\lim_{x\to\infty} f(x) = \infty$ is used to indicate that the values of $f(x)$ become large as x becomes large. Similar meanings are attached to the following symbols.

$\lim_{x\to-\infty} f(x) = \infty$, $\lim_{x\to\infty} f(x) = -\infty$, $\lim_{x\to-\infty} f(x) = -\infty$

Example: Find $\lim_{x\to\infty} x^3$ and $\lim_{x\to-\infty} x^3$.

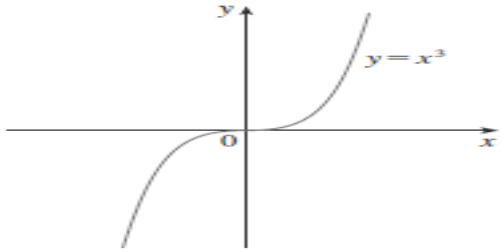
Solution: When x becomes large, x^3 also becomes large, For instance.

$10^3 = 1000$ $100^3 = 1{,}0000{,}000$ $1000^3 = 1{,}000{,}000{,}000$

In fact ,we can make x^3 as big as we like by taking x large enough.

Therefore we can write

$$\lim_{x\to\infty} x^3 = \infty$$



Similarly, when x is large negative, so is x^3. Thus $\lim_{x\to-\infty} x^3 = -\infty$

Infinite Limits

Infinite limits can also be defined in a precise way. Let us look again at the function $f(x) = \frac{1}{x}$ as $x \to o^+$ the values of f grow with out bound, that is

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \frac{1}{x} = \infty$$

And as x → 0^-, the values of $f(x) = \frac{1}{x}$ become negative, that is

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{1}{x} = -\infty$$

Example: Find $\lim_{x\to 1^+} \frac{1}{x-1}$ and $\lim_{x\to 1^-} \frac{1}{x-1}$

Geomeric solution: The graph of $f(x) = \frac{1}{x-1}$ is the graph of $f(x) = \frac{1}{x}$ shifted 1 unit to the right (see Figure 1.21) therefore $y = \frac{1}{x-1}$ be haves near 1 exactly the way $y = \frac{1}{x}$ behaves near 0.

$\lim_{x\to 1^+} \frac{1}{x-1} = \infty$ and $\lim_{x\to 1^-} \frac{1}{x-1} = -\infty$

Exercises

Determine the infinite limit.

a) $\lim_{x\to\infty} \frac{1}{2x+3}$ *b)* $\lim_{y\to-\infty} \frac{2-3y^2}{5y^2+4y}$ *c)* $\lim_{x\to 0} \frac{5}{x^5}$

d) $\lim_{x\to-\infty} \frac{1-x-x^2}{2x^3-x^2+3}$ *e)* $\lim_{x\to\infty} \frac{3x+5}{x-4}$ *f)* $\lim_{x\to 0^+} \frac{2}{x^{1/4}}$

g) $\lim_{t\to-\infty} \frac{t^2+2}{t^3+t^2-1}$ *h)* $\lim_{x\to-\infty} \frac{x+2}{\sqrt{9x+1}}$ *i)* $\lim_{x\to-\infty} (x^4 + x^5)$

j) $\lim_{u\to-\infty} \frac{-u^4+u}{(u^2-2)(2u^2-1)}$ *k)* $\lim_{x\to-\infty} \left(x + \sqrt{x^2+2x}\right)$ *l)* $\lim_{x\to-\infty} \frac{\sqrt{9x^6-x}}{x^3+1}$

m) $\lim_{x\to-\infty} \frac{\sqrt{9x^6-x}}{x^3+1}$

Continuity

Continuity of a function at a point: Let f be a function on a subset of the real numbers and let a be a point in the domain of f. Then f is continuous at a if $\lim_{x\to a} f(x) = f(a)$. i.e.

f is continuous at a if it satisfies the following three conditions:

1. $f(a)$ is defined 2. $\lim_{x\to a} f(x)$ is exist 3. $\lim_{x\to a} f(x) = f(a)$.

<u>One Sided Continuity</u>

i. A function f is continuous from the right at a number a if $\lim_{x\to a^+} f(x) = f(a)$.

ii. A function f is continuous from the left at a number a if $\lim_{x\to a^-} f(x) = f(a)$.

<u>Continuity in an interval</u>

a) f is said to be continuous in an open interval (a, b) if it is continuous at every point in this interval.

b) f is said to be continuous in the closed interval $[a, b]$ if f is continuous in (a, b), such that $\lim_{x\to a^+} f(x) = f(a)$ and $\lim_{x\to b^-} f(x) = f(b)$

<u>Discontinuity at a point</u>

The function f will be discontinuous at $x = a$ in any of the following cases:

a. $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ exist but not equal

b. $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ exist and are equal but not equalto $f(a)$.

c. $f(a)$ is not defined.

Example: 1. Find the point (s) of discontinuous of $f(x) = \frac{x^2-x-2}{x-2}$.

2. Show that the following functions are continuous or not at indicated point

a) $f(x) = \begin{cases} \frac{1}{x^2}, & if\ x \neq 0 \\ 1 & if\ \ x = 0 \end{cases}$ at $x = 0$. b) $f(x) = \begin{cases} \frac{x^2-x-2}{x-2}, & if\ x \neq 2 \\ 3 & if\ \ x = 2 \end{cases}$ at $x = 2$.

3. If $f(x) = \begin{cases} \frac{x^3+x^2-16x+20}{(x-2)^2}, & x \neq 2 \\ k, & x = 2 \end{cases}$ is continuous at $x = 2$, find the value for k.

Solution Given $f(2) = k$. Now,

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = \lim_{x\to 2} \frac{x^3+x^2-16x+20}{(x-2)^2} = \lim_{x\to 2} \frac{(x+5)(x-2)^2}{(x-2)^2} = \lim_{x\to 2}(x+5) = 7.$$

As f is continuous at $x = 2$, we have $\lim_{x\to 2} f(x) = f(2) \Longrightarrow k = 7$.

<u>Remark:</u> a) Polynomial functions, Rational functions, Root functions, Trigonometric functions, Inverse functions, Exponential functions and Logarithmic functions are continuous on their domain.

b) If $f(x)$ is continuous at $x = \boldsymbol{a}$, then $|f(x)|$ is also continuous at $x = \boldsymbol{a}$.

c) If $\lim_{x\to a} g(x) = b$ and f is continuous at b, then $\lim_{x\to a} f\big(g(x)\big) = f(b)$

Exercises

1. *Determine whether f is continuous or discontinuous at a.if f is discontinuous, determine whether f is continuous from the left at a, is continuous from the right at a.*

   a) $f(x) = \sqrt{x^2 - 13}$ $: a = -\sqrt{13}$

   b) $f(x) = \begin{cases} 3x - 4 & for\ x < 2 \\ 3x + 4 & for\ x > 2 \end{cases}$ $\quad a = 2$

   c) $f(x) = \begin{cases} \frac{|x-4|}{x-4} & for\ x \neq 4 \\ 1 & for = 4 \end{cases}$

2. *Find the values for the constant k, that makesthe following functions are continuous at $x = 0$ or continuous every where.*

   a) $f(x) = \begin{cases} \frac{\sin 3x}{x}, & x \neq 0 \\ k\ , & x = 0 \end{cases}$ *b)* $g(x) = \begin{cases} \frac{\tan kx}{x}\ , & x < 0 \\ 3x + 2k^2, & x > 0 \end{cases}$

   c) $f(x) = \begin{cases} 7x - 2, x < 1 \\ kx^2\ , x > 1 \end{cases}$ *d)* $f(x) = \begin{cases} kx^2\ , x \leq 2 \\ 2x + k, x > 2 \end{cases}$

   e) $f(x) = \begin{cases} k_1 x - k, & x \leq -1 \\ 2x^2 + 3k_1 x + k_2 & -1 \leq x \leq 1 \\ 4 & x > 1 \end{cases}$

3. *If f and g are continuous functions with $f(3) = 5$ and $\lim_{x \to 3}[2f(x) - g(x)] = 4$ then find $g(3)$.*

<u>Intermediate value theorem</u>

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, so that $f(a) \leq N \leq f(b)$ or $f(b) \leq n \leq f(a)$. Then, there exists a number c in (a, b) such that $f(c) = N$.

Example: Given $f(x) = x^3 + 2x^2 + x = 4\ on\ [-2, 1]$. Show that there exists some $c \in [-2, 1]$ such that $f(c) = 4$.

Solution: f is continuous on $[-2, 1]$ with $f(-2) = 2\ and\ f(10) = 8$. Since $2 \leq 4 \leq 8$, it follows, by the Intermediate Value Theorem that there exists $c \in [-2, 1]$ such that $f(c) = 4$. i.e. $f(c) = c^3 + 2c^2 + c + 4 = 4$. In this case we can find such c by solving

$c^3 + 2c^2 + c + 4 = 4 \Leftrightarrow c^3 + 2c^2 + c = 0 \Leftrightarrow c(c^2 + 2c + 1) = 0 \Leftrightarrow c\,(c + 1)2 = 0$, which gives either c = 0 or c = -1

Since both of these values are in [-2, 1], for this particular case we have two values in [-2, 1] with image under f equal to 4.

One of the most important applications of the Intermediate Value Theorem is given in the following theorem.

Theorem: Suppose f is continuous on a closed interval $[a, b]$ and assume that $f(a)$ *and* $f(b)$ have opposite signs. Then there is at least one $c \in (a, b)$ such that $f(c) = 0$.

Example: Show that the expression $2x^3 - 5x^2 - 10x + 5$ has at least one root between -1 and 2

Solution: Let $f(x) = 2x^3 - 5x^2 - 10x + 5$. we are lookig for a solution of the given equation , that is a number c between -1 and 2 such that $f(c) = 0$. Therefore ,we take $a = -1, b = 2$ and $N = 0$ by the theorem, we have

$$f(-1) = -4 - 5 + 10 + 5 = 6 > 0$$

$$f(2) = 16 - 20 - 20 + 5 = -19 < 0$$

Thus $f(-1) < f(2)$; that is $N = 0$ is a number between $f(-1)$ *and* $f(2)$.Now f is continuous since it is polynomial, so the intermediate value theorem says there is a number c between -1 and 2 such that $f(c) = 0$.

Exercises

Use the intermedaite value Theorem to show that there is a root of the given equation in the specified interval.

a) $x^4 + x - 3 = 0$, (1,2) *b)* $\cos x = x$, (0,1) *c)* $\ln x = e^{-x}$, (1,2)

d) $\sqrt[3]{x} = 1 - x$, (0,1) *e)* $x^3 + 2x - 1$, (0,1)

CHAPTER 2

DERIVATIVES AND APPLICATION OF DERIVATIVES

Definition of the Derivative

Let a be a number in the domain of a function f. If $\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$ exists, we call this limit the derivative of f at a and write it $f'(a)$.

Derivative of a function

Given any number x for which the limit of $f(x)$ is exists, we assign to x the number $f'(x)$. So we can regard $f'(x)$ as a new function, called the derivative of $f(x)$ and defined by

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

Example: Use definition of derivative to evaluate $f'(x)$ if

a. $f(x) = x^3$

b. $f(x) = x - \sqrt{x}$

c. $f(x) = \sqrt{x-1}$

d. $f(x) = \frac{1-x}{2+x}$.

Solution: a. Let $f(x) = x^3$, then $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{(x+h)^3-x^3}{h}$

$$= \lim_{h\to 0}\frac{x^3+3x^2h+3xh^2+h^3-x^3}{h} = \lim_{h\to 0}\frac{3x^2h+3xh^2+h^3}{h}$$

$$= \lim_{h\to 0} 3x^2 + 3xh + h^2 = 3x^2.$$

Some common alternative notations for the derivative are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

The symbols $D$and d/dx are called differentiation operators because they indicate the operation of differentiation, which is the process of calculating a derivative

Differentiability: Let a function f be defined in open interval containig the point c. The function is **differentaible** at c if and only if the derivative $f'(c)$ exists. If f is differentaile at every point of its domain we say simply that f is **differentaible**.

Example. Examine the differentiability of the function f defined by

$$f(x) = \begin{cases} 2x+3, if -3 \le x < -2 \\ x+1, \ \ if -2 \le x < 0 \\ x+2 \quad if \ \ 0 \le x \le 1 \end{cases}.$$

Solution: The only doubtful points for differentiability of $f\ (x)\ are\ x\ = -\ 2\ and\ x\ =\ 0$.

Differentiability at $x = -2$: Now

$L\,f'(-2) = \lim_{h\to 0^-} \frac{f(-2+h)-f(-2)}{h} = \lim_{h\to 0^-} \frac{2(-2+h)+3-(-2+1)}{h} = \lim_{h\to 0^-} \frac{2h}{h} = 2.$

and $R\,f'(-2) = \lim_{h\to 0^+} \frac{f(-2+h)-f(-2)}{h} = \lim_{h\to 0^+} \frac{-2+h+1-(-2+1)}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$ Thus, $Lf'(-2) \neq Rf'(-2)$. Therefore f is not differentiable at $x = -2$.

Theorem: If f differentaible at point a, then it is continuous at a.

Proof: To prove that f is continuous at a, we have to show that $\lim_{x\to a} f(x) = f(a)$.

We do this by showing that the difference $f(x) - f(a)$ approaches 0.

The given information is that f is differentiable at a, that is

$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ *ex*ists.

Now divide and multiply $f(x) - f(a)$ *by* $x - a$ (which we can do when x $\neq$ 0)

Taking the limit on both side of equation,we get

$$\lim_{x\to a}(\ f(x) - f(a)) = \lim_{x\to a} \frac{f(x) - f(a)(x-a)}{x-a}$$

$$= \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \lim_{x\to a}(x-a)$$

$$= f'(a)(0)$$

$$\Rightarrow \lim_{x\to a} f(x) = f(a)$$

Therefore, f is continuous at a.

The converse of this theorem is not true ; that is ,there are functions that are continuous at a point but may not differentaible at that point.

Example: Show that the function $f(x) = |x|$ is continuous at 0, but not differentiable at 0.

Solution: The function $f(x) = |x|$ continuous at 0, since

$$\lim_{x\to 0^-} |x| = \lim_{x\to 0^+} |x| = 0 = f(0)$$

If we compute the derivatives, we obtain

$f'(0) = \lim_{x\to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x\to 0} \frac{|x|}{x}$ using absolute value property $|x| = \begin{cases} x & for\ \ x \geq 0 \\ -x & for\ x < 0 \end{cases}$

For $x > 0$ $\lim_{x\to 0^+} \frac{|x|}{x} = \lim_{x\to 0^+} \frac{x}{x} = 1$ and

For $x < 0$ $\lim_{x\to 0^-} \frac{|x|}{x} = \lim_{x\to 0^-} \frac{-x}{x} = -1$

Thus $\lim_{x\to 0} \frac{|x|}{x}$ does not exists, because the right and the left hand side limits are not equal.

Therefore, f is not differentaible at 0.

Definition: The tangent line to the curve $y = f(x)$ at the point $p(a, f(a))$ is the line through p with slope $m = \lim_{x \to a} \frac{f(x)-f(a)}{x-a}$ provided that this limit exists. And the tangent line is given by $y - f(a) = f'(a)(x - a)$ which is called point-slope form equation of line.

Example*:* Find an equation of the tangent line to the parabola $y = x^2$ at the point $p(1,1)$.

Solution: Here we have $a = 1$ and $f(x) = x^2$, so the slope is

$$m = \lim_{x \to 1} \frac{f(x)-f(1)}{x-1}$$

$$= \lim_{x \to 1} \frac{x^2-1}{x-1}$$

$$= \lim_{x \to 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \to 1}(x + 1) = 1 + 1 = 2$$

Using the point slope form of the equation of the line, we find that an equation of the tangent line at $p(1,1)$ *is* $y - 1 = 2(x - 1)$ *or* $y = 2x - 1$

<u>Remark!</u> The slope of the line tangent to the graph of the function $y = f(x)$ *at* $(a, f(a))$ is equal to the derivative of f at a.

<u>Rules of differentiation</u>

If f and g are both differentiable, and c is constant, then the following differentiation rules are valid

a. $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x) = f'(x) \pm g'(x)$ (Sum and difference rule).

b. $\frac{d}{dx}cf(x)=c\frac{d}{dx}f(x)$ (Constant Multiple Rule)

c. $\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x) = f'(x)g(x) + f(x)g'(x)'$ (Product Rule)

d. $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}f(x)-f(x)\frac{d}{dx}g(x)}{[g(x)]^2} = \left(\frac{f}{g}\right)'(x) = \frac{f'(x)\,g(x)-f(x)g'(x)}{[g(x)]^2}$ if $g(x) \neq 0$ (Quotient rule)

<u>Derivatives of Functions</u>

<u>Derivative of constant function</u>.

If f has the constant value $f(x) = c$, then $\frac{d}{dx}f(x) = \frac{d}{dx}(c) = 0$.

<u>Power rule:</u> If n is any real number and $f(x) = x^n$, then f is defferentaible on the set of real numbers $\mathbb{R}$ and is given by $f'(x) = nx^{n-1}$

Proof: if $f(x) = x^n$ then by definition of derivative at a point, we get

$$f'(a) = \lim_{x\to a}\frac{f(x)-f(a)}{x-a} = \lim_{x\to a}\frac{x^n-a^n}{x-a} = \lim_{x\to a}\frac{(x-a)(x^{n-1}+x^{n-2}a+\cdots+xa^{n-2}+a^{n-1})}{x-a}$$
$$= \underbrace{a^{n-1}+a^{n-2}a+\cdots+aa^{n-2}+a^{n-1}}_{n\ times} = na^{n-1}.$$

Hence $f'(x) = nx^{n-1}$.

Example: Differentiate the following

a. $f(x) = \frac{1}{x^2}$ *b.* $y = \sqrt[3]{x^2}$ *c.* $g(x) = \frac{1}{\sqrt{x+2}}$

d. $f(x) = 2\pi^3$ *e.* $h(x) = (x^2+1)(x^4-1)$ *f.* $h(x) = \frac{x^2+x-2}{x^3+6}$

<u>Derivative of Trigonometric Functions</u>

Example. Using definition of derivative find derivative of $\sin x$

Solution. From the definition of a derivative, we have

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{sin(x+h)-sin(x)}{h}$$
$$= \lim_{h\to 0}\frac{sin(x)\cos(h)+\cos(x)\sin(h)-sin(x)}{h}$$
$$= \lim_{h\to 0}\left(\frac{sin(x)\cos(h)-sin(x)}{h}+\frac{\cos(x)\sin(h)}{\mathrm{h}}\right)$$
$$= \lim_{h\to 0} sin(x)\left(\frac{\cos(h)-1}{h}\right)+\lim_{h\to 0}\frac{\cos(x)\sin(h)}{\mathrm{h}}$$
$$= sin(x)\lim_{h\to 0}\left(\frac{\cos(h)-1}{h}\right)+\cos(x)\lim_{h\to 0}\frac{\sin(h)}{\mathrm{h}} = \sin x\ \cdot 0+\cos x\cdot 1 = \cos x.$$

Similarly derivative of others trig-functions are

a. $\frac{d}{dx}(\cos x) = -\sin x$ *c.* $\frac{d}{dx}(\csc x) = -\csc x\cot x$ *e.* $\frac{d}{dx}(\cot x) = -\csc^2 x$

b. $\frac{d}{dx}(\tan x) = \sec^2 x$ *d.* $\frac{d}{dx}(\sec x) = \sec x\tan x$

Example: Differentiate the following a) $f(x) = \frac{\sec x}{1-\tan x}$. b) $f(\theta) = \frac{sec\theta}{1+sec\theta}$

Solution: Here $f(x) = \frac{\sec x}{1-\tan x}$, then we have

$$\begin{aligned} f'(x) &= \frac{d}{dx}\left(\frac{\sec x}{1-\tan x}\right) \\ &= \frac{(\frac{d}{dx}\sec x)(1-\tan x) - \sec x \frac{d}{dx}(1-\tan x)}{(1-\tan x)^2} \\ &= \frac{\sec x \tan x(1-\tan x) - \sec x(-\sec^2 x)}{(1-\tan x)^2} \\ &= \frac{\sec x \tan x(1-\tan x) + \sec^3 x}{(1-\tan x)^2} \\ &= \frac{\sec x(\tan x - \tan^2 x + 1 + \tan^2 x)}{1 - 2\tan x + \tan^2 x} \\ &= \frac{\sec x(\tan x + 1)}{\sec^2 x - 2\tan x} \end{aligned}$$

<u>Derivatives of Exponential and Logarithmic Functions</u>

1. Given exponential function $f(x) = a^x$. Then $f'(x) = a^x \ln(a)$.

   In case we are using the natural exponential function $f(x) = e^x$, then $f'(x) = e^x$

2. Given logarithmic function $f(x) = \log_a x$, we have

$$f'(x) = \frac{d}{dx}(log_a x) = \frac{1}{x \ln a}$$

In case of natural logarithm $f(x) = \log_e x = \frac{lnx}{lne} = \ln x$, we have

$$f'(x) = \frac{d}{dx}(log_e x) = \frac{1}{x \ln e} = \frac{1}{x}$$

Exercises 1: Differentiate the following

1. $f(x) = 3x^2 - 2\cos x$ *2.* $h(x) = csc\, x + e^x \cot x$ 3. $y = sec^2 x$

4. $f(x) = \sin x + \frac{1}{2}\cot x$ *5.* $y = e^u(\cos u + cu)$ *6.* $y = \frac{1}{x} - \frac{1}{\sin x}$

7. $f(t) = t^3 \cos t$ *8.* $y = \frac{x}{4-\tan x}$ *9.* $f(x) = (x^2 + 3x + 1)\sin x$

10. $f(t) = \sqrt{t}\sin t$ *11.* $y = \frac{1+\sin x}{x+\cos x}$ *12.* $y = 2\sin x \cos x$

13. $y = 2\csc x + 4\tan x$ *14 .* $f(\theta) = \frac{\sec\theta}{1+\sec\theta}$ *15.* $y = \ln x \cos(x)$

16. $y = \frac{1-\sec x}{\tan x}$ *17.* $y = \log_3 x \csc x$ *18.* $y = \frac{x\sin x}{1-\cos x}$

19. $y = \frac{\sin x}{x^2}$ *20 .* $y = x^2 \sin x \tan x$ *21.* $f(x) = \frac{4x+3}{2x\sin x}$

2. Find an equation of the tangent line to the curve at the given point.

a) $y = sec\, x, (\pi/3, 3)$ *b)* $y = e^x \cos x, \ (0,1)$

c). $y = x + \cos x, \ (0,1)$ *d)* $y = \frac{1}{\sin x + \cos x}, \ (0,1)$

Derivatives of Hyperbolic Functions

Definition: $\sin hx = \frac{e^x - e^{-x}}{2}$ $\csc hx = \frac{1}{\sin hx}$

$\cos hx = \frac{e^x + e^{-x}}{2}$ $\sec hx = \frac{1}{\cos hx}$

$\tan hx = \frac{\sin hx}{\cos hx}$ $\cot hx = \frac{1}{\tan hx}$

Their derivative are given by:

1. $\frac{d}{dx}(\sin hx) = \cos \text{hx}$
2. $\frac{d}{dx}(\cos \text{hx}) = \sin hx$
3. $\frac{d}{dx}(\tan hx) = \sec \text{h}^2 \text{x}$
4. $\frac{d}{dx}(\csc hx) = -\csc hx \cot hx$
5. $\frac{d}{dx}(\sec hx) = \sec hx \tan hx$
6. $\frac{dy}{dx}(\cot hx) = -\csc h^2 x$

Proof : 1) $\frac{d}{dx}(sin\, hx) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{e^x - (-e^{-x})}{2} = \frac{e^x + e^{-x})}{2} = cos\, hx$

2) $\frac{d}{dx}(cos\, hx) = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) = \frac{e^x - e^{-x}}{2} = sin\, hx$

The remaining proof are left as exercise.

The Chain Rule

If $f\ and\ g$ are both differentiable and $F = fog$ is the composite function defined by $F(x) = f(g(x))$, the F is differentiable and F' is given by the product $F'(x) = f'\big(g(x)\big)g'(x)$.

In Leibniz notation, if $y = f(u)\ and\ u = g(x)$ are both differentiable functions, then $\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$

Note: In using the chain rule we work from the outside to the inside.

Example 1: Find derivative of the following.

a) $f(x) = \sin^2 x$ b) $y = 4^{x^2}$ c) $y = (x^3 - 1)^{99}(x^2 - 10)^4$ d) $f(x) = \ln(x^3 + 3x - 4$

Solution: a). Note that $f(x) = \sin^2 x = (\sin x)^2$. Here the outer function is the squaring function and the inner function is the sine function. So $\frac{dy}{dx} = \frac{d}{dx}(\sin x)^2 = 2 \sin x \cos x = \sin 2x$

b) $y = 4^{x^2} \Rightarrow \frac{d}{dx}\left(4^{x^2}\right) = 4^{x^2} \ln 4 \frac{d}{dx}(x^2) = 4^{x^2}(2x) \ln 4.$

c) $\frac{d}{dx}[(x^3-1)^{99}(x^2-10)^4] = (x^2-10)^4\frac{d}{dx}(x^3-1)^{99} + (x^3-1)^{99}\frac{d}{dx}(x^2-10)^4 =$

$297x^2(x^3-1)^{98}(x^2-10)^4 + 8x(x^3-1)^{99}(x^2-10)^3$

d) $f'(x) = \frac{1}{x^3+3x-4}\frac{d(x^3+3x-4)}{dx} = \frac{3x^2+3}{x^3+3x-4}$.

Exercises

1. *Find the derivative of the function*

a. $f(x) = x^{7/6} - x^{-7/6}$

b. $f(x) = \sqrt{2x^2+3x-1}$

c. $f(x) = \frac{\sqrt{x^3-1}}{x^2}$

d) $f(x) = \sqrt[3]{x+\tan x}$

e) $y = a^3 + \cos^3 x$

f) $g(t) = (6t^2-5)^3(t^2-2)^4$

g) $g(t) = \left(\frac{t-2}{2t+1}\right)^{10}$

h) $f(x) = \sqrt{x+\sqrt{x+\sqrt{x}}}$

i) $f(x) = \sqrt{\tan\sqrt{x}}$

2. *Find an equation of the tangent line to the graph of f at the given point*

a. $f(x) = \frac{2}{1+e^{-x}}$ *at the point (0,1)*

b. $f(x) = x\sqrt{2-x^2}$ *, find* $f'(x)$

3. *If* $F(x) = f(g(x)$, *where* $f(-2) = 8, f'(-2) = 4, f'(5) = 32$ *and* $g'(5) = 6$, $g(5) = -2$ *find* $F'(5)$

4. *If* $F(x) = f\big(g(x)\big)$, *where* $f(-2) = 8, f'(-2) = 4, f'(5) = 3$ $g(5) = -2$ *and* $g'(5) = 6$ *find* $F'(5)$.

5. *If* $g(x) = \sqrt{4+3f(x)}$ *where* $f(1) = 7$ *and* $f'(1) = 4$. *find* $g'(1)$

<u>Higher order derivatives</u>

If f is a differentiable function, then its derivative f' is also a function, so $f'(x)$ may have a derivative of its own, denoted by $(f')'(x) = f''(x) = D^2(x)$. This new function f'' is called the **second derivative** of f. Again the derivative of this second derivative is yield another new function which is called **third derivative of f** which is $(f'')'(x) = f'''(x) = D^3f(x)$. The process can be continued. In general, the n^{th}derivative of f is denoted by $f^{(n)}$ and is obtained from f by differentiating n times. And write as

$$y^{(n)} = f^{(n)}(x) = \frac{d^ny}{dx^n} = D^nf(x)$$

Example 1: If $f(x) = x^4 - x^3 + x^2 - x + 1$. Find the successive derivative of.

Solution: $f'(x) = 4x^3 - 3x^2 + 2x - 1 + 0$

$f''(x) = 12x^2 - 6x + 2 - 0$

$f'''(x) = 24x - 6 + 0$

$f^{(4)}(x) = 24$

$f^{(5)}(x) = 0$

.

.

$f^{(n)}(x) = 0 \qquad (n \geq 5)$

2. Find a formula for the nth derivative of $f(x) = \frac{1}{1-x}$.

Solution: By applying the quotient rule, $f'(x) = \frac{1}{(1-x)^2}$, $f''(x) = \frac{2}{(1-x)^3}$, $f'''(x) = \frac{6}{(1-x)^4}$.

So by proceeding in this way we obtain$f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$

Exercise: 1. Evaluate a) $D^4(x^3 + 2x + 8$ b) $D^{128}(\sin x)$ c) $D^{27}(\cos x)$ d) $D^4(e^{2x})$

e) $y'''(0)$, *where* $y = 4x^4 + 2x^3 + 3$ *f)* $\left.\frac{d^4y}{dy^4}\right|_{x=1}$, *where* $y = \frac{6}{x^4}$

2. *Find the higher order or n^th derivatives of the following*

a) $f(x) = \frac{1}{x}$ *b)* $f(x) = xe^x$ *c)* $f(x) = e^{ax}$ *d)* $f(x) = \ln x$

<u>Implicit differentiation</u>

Implicit differentiation is a technique for differentiating functions that are not given in the usual form$y = f(x)$. In general, although an equation of the form

$$F(x, y) = 0 \qquad (1)$$

Usually defines *y* as a function of *x* implicitly. To find $\boldsymbol{\frac{dy}{dx}}$, we may differentiate both sides of (1) with respect to *x* and solve for $\boldsymbol{\frac{dy}{dx}}$ (in terms of *x* and *y*). This method is called ***implicitdifferentiation.***

This implies that we can have two methods to find the derivative of such functions:

1. Rewrite the given functions in the form of $\boldsymbol{y = f(x)}$ (if it is possible).

2. Differentiate both side with respect to independent variable, x, and simplifies the solution for $\frac{dy}{dx}$. (NB. Here we can use the Chain Rule & Power Rule by treating y as a function of x).

Example Find the slope of the line tangent to the circle $x^2 + y^2 = 4$ at the point $(\sqrt{2}, \sqrt{2})$.

Soln. First rewrite the equation of the circle in the form $y = f(x)$ and we get; $y = \pm\sqrt{4 - x^2}$. But this doesn't give a function since certain value of x gives two values of y. Note that the point $(\sqrt{2}, \sqrt{2})$ lies on the upper semi-circle. To consider the required slope, we take $y = \sqrt{4 - x^2}$. Secondly, differentiate both side with respect x by treating $y = f(x)$ as a function of x, we get

$$\frac{dy}{dx} = \frac{d}{dx}\left(\sqrt{4 - x^2}\right) = \frac{1}{2\sqrt{4 - x^2}} \cdot \frac{d}{dx}(4 - x^2) = \frac{-x}{\sqrt{4 - x^2}}.$$

The slope of the tangent at $(\sqrt{2}, \sqrt{2})$ is $\frac{dy}{dx}|_{x=\sqrt{2}} = \frac{-\sqrt{2}}{\sqrt{2}} = -1$.

Example 1: Find $\boldsymbol{\frac{dy}{dx}}$given that a) $x^3 + 4xy^2 - 7 = y^3$ b) $e^x + e^y = e^{x+y}$ c) $ylnx = xe^y - 1$

d) $y = tan(x + y)$ *e)* $sin(x + y) = y\, cos\, x$ $f)$ $e^y = x^2 + y^2$

Soln. a) Since to solve y as a function of x explicitly, we assume that y is a function of x *and* differentiate both sides of the given equation with respect to x, we get

$$\frac{d}{dx}(x^3 + 4xy^2 - 7) = \frac{d}{dx}(y^3) \Leftrightarrow 3x^2 + 4y^2 + 8xy\frac{dy}{dx} = 3y^2\frac{dy}{dx}$$

Then solving for $\frac{dy}{dx}$, we get

$$3x^2 + 4y^2 = (3x^2 - 8xy)\frac{dy}{dx} \Longrightarrow \frac{dy}{dx} = \frac{3x^2 + 4y^2}{3x^2 - 8xy}$$

2. Using implicit differentiation find $\frac{d^2y}{dx^2}$ *if* $x^2 + y^2 = 1$

Solution: Differentiating both sides of $x^2 + y^2 = 1$ with respect to x implicitly yields

$$2x + 2y\frac{dy}{dx} = 0$$

$$\Longrightarrow \frac{dy}{dx} = \frac{-x}{y}$$

Again applying implicit differentiation for the second time, we get

$$\frac{d^2y}{dx^2} = \frac{(-1)y - (-x)y'}{y^2} \qquad (y' = \frac{-x}{y})$$

$$= \frac{-y + x\frac{-x}{y}}{y^2} = -\frac{1}{y^3} \qquad (x^2 + y^2 = 1)$$

3. Find the equation of tangent line to the functions $x^3 + y^3 = 3xy$ *at* $\left(\frac{3}{2}, \frac{3}{2}\right)$

Exercises

1. *Find* $\frac{dy}{dx}$ *by implicit differentiation.*

   a) $x^2 + y^2 = 68$

   b) $x^3y^2 - 5x^2y + x = 1$

   c) $x^2 = \frac{x+y}{x-y}$

   d) $\sqrt{xy} = 1 + x^2y$

   e) $e^y cosx = 1 + sin(xy)$

   f) $x^2y + 3xy^3 - x = 3$

   g) $e^{x/y} = x - y$

   h) $ysin(x^2) = xsin\ (y^2)$

   i) $\sqrt{x+y} = \frac{y}{1+x^2}$

2. For each of the following curves represented by the given equations, find the slope at the indicated point. *a)* $2y^3 + y^2 - x = 0,\ (3,1)$ *b)* $xsin\ y\ +\ cosy^2 = 1$ *at* $(1,0)$

   c) $ln\ y\ =\ 2y^2 - x + 1,\ \ (3,1)$ *d)* $ln(x + sin\ y) = x^2 + 2e^y - 3,\ \ (1,0)$

<u>Logarithmic differentiation</u>

Logarithmic differentiation is a powerful technique to differentiate functions of the form $f(x) = (u(x))^{v(x)}$, where both f and u need to be positive functions for this technique to make sense.

<u>Steps in Logarithmic Differentiation</u>

1. Take natural logarithms both sides of an equation $y\ =\ f(x)$ and use the Laws of Logarithms and simplify.
2. Differentiate implicitly with respect to x.
3. Solve the resulting equation for $\frac{dy}{dx}$

Example. Find the derivative of the following functions. a) $y = (2x^4 + 1)^{tanx}$ b) $y = (\ln x)^x$

Solution: a) Take natural logarithms both sides of a given equation as

$\ln y = \ln[(2x^4 + 1)^{tanx}] \Longrightarrow \ln y = tanx \ln(2x^4 + 1)$.

Differentiate implicitly wrt x; $\frac{d \ln y}{dx} = \frac{d(tanx \ln(2x^4+1))}{dx} \Longrightarrow \frac{1}{y}\frac{dy}{dx} = sec^2x \ln(2x^4 + 1) + \tan x \left(\frac{8x^3}{2x^4+1}\right)$

Solve the resulting equation for $\frac{dy}{dx}$

$\frac{dy}{dx} = y\left[sec^2x\ln(2x^4+1) + \tan x\left(\frac{8x^3}{2x^4+1}\right)\right] = (2x^4+1)^{tanx}\left[sec^2x\ln(2x^4+1) + \tan x\left(\frac{8x^3}{2x^4+1}\right)\right]$.

b) Take natural logarithms both sides of a given equation; $\ln y = \ln[(\ln x)^x] \Longrightarrow \ln y = x\ln(\ln x)$.

Differentiate implicitly wrt x; $\frac{d\ln y}{dx} = \frac{d(x\ln(\ln x))}{dx} \Longrightarrow \frac{1}{y}\frac{dy}{dx} = \ln(\ln x) + \frac{x}{\ln(x)}\frac{1}{x} = \ln(\ln x) + \frac{1}{\ln(x)}$

Solve the resulting equation for $\frac{dy}{dx}$; $\frac{dy}{dx} = y\left[\ln(\ln x) + \frac{1}{\ln(x)}\right] = (\ln x)^x\left[\ln(\ln x) + \frac{1}{\ln(x)}\right]$.

Exercise. 1. Find $\frac{dy}{dx}$ for the equation a) $y = \sqrt[x]{x}$ b) $y = (\sin x)^{\cos x}$ c) $y = x^{\csc x}$ d) $y = \frac{x\sqrt{2x+1}}{e^x sin^2x}$

2. If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\ln x}{(1+\ln x)^2}$.

<u>Inverse functions and their derivatives</u>

Definition. Let f and g are functions such that the domain of g is the range f and $f(x) = y$ if and only if $g(y) = x$ for all x in the domain of f and for all y in the range of f. then g is the inverse function of f and denoted by $f(x) = g^{-1}(x)$.

Theorem: A function f is said to be invertible iff it is one to one.

Steps to find inverse of a function

Step 1. Write $y = f(x)$. step 2. Interchange place x *and* y. step 3. Solve for y in terms of x in step 2. Step 4. Write $f^{-1}(x)$ for y.

Example. Find the inverse of *a)* $f(x) = 3x + 7$ *b)* $f(x) = \sqrt{e^x + 1}$ *c)* $f(x) = ln(3x + 2)$

d) $f(x) = x^3 - 9$ *e)* $f(x) = \sqrt{\frac{x+2}{x}}, x > 0$

Soln: $step\ 1. y = 3x + 7\ \ step\ 2.\ x = 3y + 7.\ step\ 3\ y = \frac{x}{3} - \frac{7}{3}. step\ 4 f^{-1}(x) = \frac{x}{3} - \frac{7}{3}$

<u>Derivative of Inverse function</u>

Theorem. Suppose that f has an inverse and it is continuous on an open interval containing a assume also $f'(a)$ exist and $f'(a) \neq 0$, and $f(a) = c$, then $(f^{-1})'(c)$ exist and $(f^{-1})'(c) = \frac{1}{f'(a)}$

Example: Let $f(x) = x^7 + 8x^3 + 4x - 2$, then find $(f^{-1})'(-2)$.

Solution: Let us find the value of a for which $f(a) = -2 \Rightarrow f(0) = -2$, so $a = 0$. Since

$f'(x) = 7x^6 + 24x^2 + 4$. It follow that $f'(0) = 4 \Rightarrow (f^{-1})'(-2) = \frac{1}{f'(0)} = 1/4$.

Exercise. a) Let $f(x) = x^5 + 2$, then find $(f^{-1})'(2)$ b) Let $f(x) = x + sin\, x$,then find $(f^{-1})'(0)$

Note: Putting $y = f(x) \Longrightarrow (f^{-1})(y) = x$ then find $(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{\frac{dy}{dx}} = \frac{dx}{dy}$

Example: Find $(f^{-1})'(y)$ if a) $f(x) = \sqrt{e^x + 1}$ b) $f(x) = ln(x^3 + 2)$ c) $f(x) = \frac{e^{2x}+1}{e^x+1}$

Solution: a) $(x) = \sqrt{e^x + 1} \Rightarrow f'(x) = \frac{e^x}{2\sqrt{e^x+1}}$. Since $(f^{-1})'(y) = \frac{1}{f'(x)}$, then

$(f^{-1})'(y) = \frac{1}{\frac{e^x}{2\sqrt{e^x+1}}} = \frac{2\sqrt{e^x+1}}{e^x}$.

Definition: If f has an inverse and the value of f^{-1} varies over an interval on which f has a nonzero derivative, then $f^{-1}$is differentiable and the derivative is given by the formula:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

This formula is an immediate consequence of the definition of an inverse function and the chain rule:

$$f\left(f^{-1}(x)\right) = x \Longrightarrow \frac{d}{dx}\left[f\left(f^{-1}(x)\right)\right] = \frac{dx}{dx} \Longrightarrow f'\left(f^{-1}(x)\right) * \left(f^{-1}\right)'(x)\right) = 1$$

$$\Longrightarrow (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

Example: Find the inverse and its derivative of (a) $f(x) = x^2 \quad on\ (0, \infty)$ $\quad (b)\ f(x) = log^{(x+1)}$

Sol<u>n</u>:(a) we have $f^{-1}(x) = \sqrt{x}$ and also $f'(x) = 2x$ so that $f'\left(f^{-1}(x)\right) = 2\sqrt{x}$. thus

$$(f^{-1})'(x) = \frac{1}{f'\left(f^{-1}(x)\right)} = \frac{1}{2\sqrt{x}}$$

b) General way to starting this is $f\left(f^{-1}(x)\right) = x$ for all x. For this problem,

$f\left(f^{-1}(x+1)\right) = log^{(f^{-1}(x)+1)} = x \Longrightarrow 10^x = f^{-1}(x) + 1 \Longrightarrow f^{-1}(x) = 10^x - 1$. Thus,

$f'(x) = \frac{1}{(x+1)ln10} \Longrightarrow f'\left(f^{-1}(x)\right) = \frac{1}{(10^x-1+1)ln10} = \frac{1}{10^x ln10}$ hence using the formula

$$(f^{-1})'(x) = \frac{1}{f'\left(f^{-1}(x)\right)} = \frac{1}{\frac{1}{10^x ln10}} = 10^x ln10.$$

Finding the inverse of a logarithmic function.

If the logarithmic function is one to one, its inverse exists. The inverse of a logarithmic function is an exponential. Steps to find the inverse of a logarithm

Step 1: replace the function notation $f(x)$ by y.

Step 2: switch the roles of x and y.

Step 3: Isolate the **log expression** on one side of the equation.

Step 4: Convert the log equation into its equivalent exponential.

Step 5: Solve for the exponential equation for y to get the inverse. Then replace y by $f^{-1}(x)$.

Ex. Find the inverse and its derivative of the function

a) $f(x) = log_5^{(2x-1)} - 7$ b) $f(x) = \ln(x-7) + 3$

Soln: a) Let begin by changing the $f(x)$ int y, and swapping the x and y.

$$f(x) = y = log_5^{(2x-1)} - 7 \Longrightarrow x = log_5^{(2y-1)} - 7 \Rightarrow x + 7 = log_5^{(2y-1)}$$

$5^{(x+7)} = 2y - 1 \Longrightarrow y = \frac{5^{(x+7)}+1}{2} \Longrightarrow f^{-1}(x) = \frac{5^{(x+7)}+1}{2}$. Then, $(f^{-1})'(x) = \frac{5^{(x+7)}\ln 5}{2}$

Derivatives of Inverse trigonometric functions

By using implicit differentiation, we can find the derivatives of the inverse trigonometric functions, assuming that these functions are differentiable. Now, recall that the definition of the arcsine function: $\boldsymbol{y = sin^{-1}x}$means $siny = x$ and $\frac{-\pi}{2} \leq y \leq \frac{\pi}{2}$. Differentiating$\boldsymbol{siny = x}$ implicitly with respect to x, we obtain $\boldsymbol{cosy\frac{dy}{dx} = 1} \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$. $\boldsymbol{\cos y \geq 0}$, since$\frac{-\pi}{2} \leq y \leq \frac{\pi}{2}$, so $\cos y = \sqrt{1-\sin^2 y} = \sqrt{1-\mathrm{x}^2}$. Hence, $\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\mathrm{x}^2}}$.

Therefore, $\frac{d}{dx}(sin^{-1}x) = \frac{1}{\sqrt{1-\mathrm{x}^2}}$.

Similarly, if $y = tan^{-1}x$ we mean that $tany = x$. Then differentiating implicitly with respect to x, we have

$$sec^2y\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{sec^2y} = \frac{1}{1+tan^2y} = \frac{1}{1+x^2}.$$

Hence, $\frac{d}{dx}(tan^{-1}x) = \frac{1}{1+x^2}$. and the derivative of others inverse trigonometric functions are derived in similarly manners and we get

$\frac{d}{dx}(sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$ $for\ y \in \left(0, \frac{\pi}{2}\right] \cup \left(\pi, \frac{3\pi}{2}\right]$ and $\frac{d}{dx}(sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$ $for\ y \in \left(\frac{\pi}{2},\ \pi\right]$

$\frac{d}{dx}(cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$, $\frac{d}{dx}(csc^{-1}x) = -\frac{1}{x\sqrt{x^2-1}}$, and $\frac{d}{dx}(cot^{-1}x) = -\frac{1}{1+x^2}$.

Example. Find the derivative of the function;

(a) $y = sin^{-1}(2x-1), x\epsilon(0,1)$ (b) $y = cos^{-1}(e^{2x})$

(c) $y = cot^{-1}\sqrt{x^2+2} + cot^{-1}\frac{1}{x}$ d) $y = x\tan^{-1}(\cos x)$.

Soln: (a) $\frac{dy}{dx} = \frac{d}{dx}\left(sin^{-1}(2x-1)\right) = \frac{1}{\sqrt{1-(2x-1)^2}}\frac{d}{dx}(2x-1) = \frac{2}{\sqrt{4x-4x^2}} = \frac{1}{\sqrt{x-x^2}}, x\epsilon(0,1)$.

Derivative of Hyperbolic Functions

$\frac{d}{dx}(\sinh x) = \frac{d}{dx}\left(\frac{e^x-e^{-x}}{2}\right) = \frac{e^x+e^{-x}}{2} = \cosh x$, $\frac{d}{dx}(\cosh x) = \sinh x$, $\frac{d}{dx}(\tanh x) = \text{sech}^2 x$,

$\frac{d}{dx}(csch\, x) = -\, csch\, x\, coth\, x$, $\frac{d}{dx}(sech\, x) = -sech\, x\, tanh\, x$, $\frac{d}{dx}(coth\, x) = -csch^2 x$

Exercises. Find derivative of *(a)* $y = x\, cosh\,(ln\, x)$ *(b)*$y = e^{sinh\, 2x}$

(c) $y = coth\sqrt{1+x^2}$ *d)* $y = e^x sech\, x$

Inverse Hyperbolic Functions

- Let $y = sinh^{-1}x$, then $x = \sinh y$ and from hyperbolic identity $\cosh^2 y - \sinh^2 y = 1$ with $coshy > 0, \cosh y = \sqrt{1+\sinh^2 y} = \sqrt{1+x^2}$ and $e^y = \cosh y + \sinh y = x + \sqrt{1+x^2} \Rightarrow y = \ln\left(x+\sqrt{1+x^2}\right). \forall x \in |R$.
- Let $y = cosh^{-1}x$, then $x = coshy\ \ and\ y \geq 0$ so $sinhy = \sqrt{\cosh^2 y - 1} = \sqrt{x^2-1}$ and $e^y = \cosh y + \sinh y = x + \sqrt{x^2-1} \Rightarrow y = \ln\left(x+\sqrt{x^2-1}\right)\ \ x \geq 1$.
- Let, $y = sech^{-1}x$. Then $x = sechy = \frac{2}{e^y+e^{-y}} \Rightarrow xe^y + xe^{-y} = 2 \Rightarrow x(e^y)^2 + x - 2e^{-y} = 0 \Leftrightarrow e^y = \frac{1\pm\sqrt{1-x^2}}{x}$. But $y > 0 \Rightarrow e^y > 1$. This rules out the minus sign because $\frac{1-\sqrt{1-x^2}}{x} > 1 \Leftrightarrow 1-\sqrt{1-x^2} > x \Leftrightarrow 1-x > \sqrt{1-x^2} \Leftrightarrow 1-2x+x^2 > 1-x^2 \Leftrightarrow x^2 > x$, but $x = \text{sech}\, y \leq 1$. Thus, $e^y = \frac{1+\sqrt{1-x^2}}{x} \Longrightarrow sech^{-1}x = \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)$.

Similarly the inverse of other Hyperbolic function is

$$tanh^{-1}x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right), \quad csch^{-1}x = \ln\left(\frac{1}{x} + \frac{\sqrt{x^2+1}}{|x|}\right), \quad coth^{-1}x = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$$

Derivative of inverse Hyperbolic Functions

Let $y = csch^{-1}x \Rightarrow \operatorname{csch} y = x \Rightarrow \frac{d}{dx}(\operatorname{csch} y) = -\operatorname{csch} y \coth y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\operatorname{csch} y \coth y}$

$\coth y = \pm\sqrt{csch^2 y + 1} = \pm\sqrt{x^2+1}$. If $x > 0$, then $\coth y > 0$, so $\coth y = \sqrt{x^2+1}$. If $x < 0$, then $\coth y < 0$, so $\coth y = -\sqrt{x^2+1}$. In either case we have $\frac{dy}{dx} = -\frac{1}{\operatorname{csch} y \coth y} = \frac{1}{|x|\sqrt{x^2+1}}$,

Similarly, derivative of other inverse Hyperbolic Functions are

$\frac{d}{dx}(sinh^{-1}x) = \frac{1}{\sqrt{x^2+1}}$, $\frac{d}{dx}(cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}}$ $\frac{d}{dx}(tanh^{-1}x) = \frac{1}{1-x^2}$

$\frac{d}{dx}(sech^{-1}x) = -\frac{1}{x\sqrt{1-x^2}}$, $\frac{d}{dx}(coth^{-1}x) = \frac{1}{1-x^2}$

Example: Find derivative of $a)\ tanh^{-1}(\sin x)$ $b)\ coth^{-1}\sqrt{x^2+1}$ c) $x sinh^{-1}\left(\frac{x}{3}\right) - \sqrt{9+x^2}$

Soln: $\frac{d}{dx}\left(tanh^{-1}(\sin x)\right) = \frac{1}{1-sin^2 x}\frac{d}{dx}(\sin x) = \frac{\cos x}{1-sin^2 x} = \frac{\cos x}{cos^2 x} = \frac{1}{\cos x} = \sec x.$

Application of derivative

Extreme Values of Functions

- ☞ A function f has a **local maximum** (or **relative maximum**) at c if $f(c) \geq f(x)$ when x is near c. [This means that for all x in some open interval containing c.] Similarly, f has a **local minimum** at c if $f(c) \leq f(x)$ when x is near c.
- ☞ A function f has an **absolute maximum** (or **global maximum**) at c if f(c) $\geq$ f(x) for all x in D , where D is the domain of f. The number f(c) is called the maximum value of f on D . Similarly, f has an **absolute minimum** at c if f(c) $\leq$ f(x) for all x in D and the number f(c) is called the minimum value of f on D. The maximum and minimum values of f are called the extreme values of f.

Example: If $f(x) = x^2$, then $f(x) \geq 0$ for all x. therefore, $f(0) = 0$ is the absolute and also local minimum value of f. This corresponds to the fact that the origin is the lowest point on the parabola $y = x^2$. However, there is no highest point on the parabola and so this function has no maximum extreme value.

Example: Find local and absolute extreme value of a) $f(x) = \cos x$ b) $f(x) = x^3$

The Extreme Value Theorem:

If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

This theorem says that a continuous function on a closed interval has a maximum value and a minimum value, but it does not tell us how to find these extreme values.

Fermat's Theorem:

If f has a local maximum or minimum at c, and if $f'(c)$ exists, then $f'(c) = 0$.

Caution: The converse of Fermat's theorem is false in general. However, it suggests that we should start looking for extreme values at numbers where the derivative is zero or does not exist.

A critical number of a function f is a number **c** in the domain of f such that either $\boldsymbol{f'(c) = 0}$ or $f'(c)$ does not exist.

Exercises. Find the critical numbers of the following

a) $f(x) = \frac{x}{1+x^2}$ *b)* $f(x) = x\ln x$ *c)* $f(x) = xe^{2x}$ *d)* $f(x) = \sqrt{x^2 - 1}$.

Rephrase Fermat's Theorem

If f has a local maximum or minimum at c, then c is a critical number of f

The Closed Interval Method

To find the **absolute Extreme values** of a continuous function f on a closed interval [a,b]:

1. Find the values of f at the critical numbers of f in (a, b)
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example: Let $f(x) = x - x^3$. Find the extreme values of f on [0,1] and determine at which numbers in [0,1] they occur.

Solution: First we find the critical points of f. We know f has extreme values because it is continuous on [0,1], as f is differentiable.

$$f'(c) = 1 - 3c^2 = 0$$

$c = \pm\frac{\sqrt{3}}{3}$. Since $-\frac{\sqrt{3}}{3}$ is not in the interval [0,1], the only critical number in our focus is $\frac{\sqrt{3}}{3}$.

We compute the corresponding value of f,

$$f(0) = 0, f(1) = 0, f\left(\frac{\sqrt{3}}{3}\right) = \frac{2}{9}\sqrt{3}$$

Consequently, the minimum value of f on [0,1] is 0 and it occurs at 0 and 1. The maximum value of f on [0,1] is $\frac{2}{9}\sqrt{3}$ and it occurs at $\frac{\sqrt{3}}{3}$.

Exercise. *Find the absolute maximum and the absolute minimum values of f on the given interval*

a) $f(x) = x^{4/5}(x-4)^2,\ [-2, 2\,]$

b) $f(x) = 3x^{2/3} - 2x,\ \ [-1, 1]$

c) $f(x) = x\sqrt{9 - x^2}\qquad [-2, 2]$

d) $f(x) = x^3 - 3x + 1\ \ ,[0,3]$

e) $f(x) = e^x + e^{-2x},\ [0, 1]$

f) $f(x) = 3x^2 - 12x + 5\qquad [0, 3]$

g) $f(x) = \sin x\ + \cos x\quad ,[0, \frac{\pi}{3}]$

<u>Rolle's Theorem</u> Let f be a function such that

- f is continuous on the closed interval $[a, b]$.
- f is differentiable on the open interval (a, b).
- $f(a)\ =\ f(b)$

Then there is a number c in (a, b) such that $f'(c)\ =\ 0$

Exercise: 1. If the function $\boldsymbol{f(x) = ax^3 + bx^2 + 11x - 6}$ satisfies conditions of Rolle's Theorem in $[1\,, 3]$ and $\boldsymbol{f'\left(2 + \frac{1}{\sqrt{3}}\right) = 0}$, then find values of **a** and **b** respectively. **Ans** $a = 1\ and\ b = -6$.

2. If $f(x)\ =\ \boldsymbol{x}\sqrt{\boldsymbol{x}+4}$ on $[-4, 0]$, then find value(s) of $c \in (-4, 0)$ that satisfy the conclusion of Roll's theorem. Ans. $c = -8/3$.

<u>The Mean Value Theorem (MVT)</u>: Let f be a function that satisfies:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a,b).

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Example. Find a number c that satisfies MVT for the function

a) $f(x) = x^3 - x \; on \; [0, 2]$. b) $f(x) = -3 + \sqrt{x}, \quad a = 0, \; b = 1$.

Solution: (a) The function $f(x) = x^3 - x$ is continuous and differentiable on the given interval. Then we can find a number c in $(0, 2)$ such that

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} \Longrightarrow 3c^2 - 1 = \frac{6 - 2}{2} \Longrightarrow 3c^2 = 3 \Longrightarrow c = \pm 1$$

Then since -1 is not element of $(0, 2)$, a number c that satisfies MVT is $c = 1$.

Monotonic Functions

Increasing/Decreasing Test

- If $f'(x) > 0$ for all x on the interval I, then the function f is strictly increasing on I.
- If $f'(x) < 0$ for all x on the interval I, then the function f is strictly decreasing on I.
- If $f'(x) = 0$ for all x on the interval I, then the function f is constant on I.

Example: Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and on which f is decreasing.

Solution: The function f is everywhere continuous and differentiable. The first derivative $f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2)$ is zero at $x = -2$ and $x = 2$. These critical points subdivide the domain of f to create non overlapping open intervals $(-\infty, -2), (-2, 2), and \; (2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f' at a convenient point in each subinterval.

Intervals	$(-\infty, -2)$	$(-2, 2)$,	$(2, \infty)$
Sign of f'	$+ + + + +$	$- - - - - -$	$+ + + + +$
Behavior of f	Increasing	decreasing	Increasing

The First Derivative Test for local extremum

Suppose that c is a critical number of a continuous function f.

a) If f' changes from positive to negative at c, then f has a local maximum at c.

b) If f' changes from negative to positive at c, then f has a local minimum at c.

c) If f' does not change sign at c (for example, if f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c.

Example. i) Find the intervals on which f is increasing or decreasing.

ii) Find the relative maximum and minimum values of the following function.

a) $f(x) = \frac{1}{4}x^3 - 3x$ b) $f(x) = x\ln x$ c) $f(x) = xe^x$ d) $f(x) = \frac{\ln x}{x}$ e) $f(x) = \frac{x}{x^2+2}$

Soln: a) $f(x) = \frac{1}{4}x^3 - 3x$, $f'(x) = \frac{3}{4}x^2 - 3 = \frac{3}{4}(x^2 - 4) = \frac{3}{4}(x-2)(x+2)$

				-2			2			
$x+2$		-	-	0	+		+		+	
$x-2$	-	-	-	-		-	0	+		+
$f(x) = \frac{3}{4}(x+2)(x-2)$	+		+	0	-		-	0	+ +	+

Then, $f(x)$ is increase on $(-\infty, 2) and\ (2, \infty)$ and decrease on $(-2, 2)$ and also

$f'(x)$ Changes sign from positive to negative at -2, so $f(-2) = 4$ is relative maximum value.

$f'(x)$ Changes sign from negative to positive at 2, so $f(2) = -4$ is relative minimum of f.

The Second Derivative Test: Suppose $f''$is continuous near c.

(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

(b) f $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

Exercise: Let $a)\ f(x) = 4x^2 + 3x$ $b)\ f(x) = \frac{x^2}{x-1}$, find the local extremum values

Ans. a) f has local min at $x = -3/8$. b) f has a local max. at $x = 0$ and local min. at $x = 2$

<u>Concavity and Inflection points</u>

Definition: If the graph of f lies above all of its tangents on an interval I, then it is called **concave upward** on I. If the graph of f lies below all of its tangents on I, then it is called **concave downward** on I.

Concavity Test

a) If $f''(x) > 0$ for all x in , then the graph of f is concave upward on I.

b) If $f''(x) < 0$ for all x in I, then the graph of f is concave downward on I.

Definition: A point p on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at p.

To find inflection point we can follow the following steps**;**

1. Find the value of c for which $f''(c) = 0$ or $f''(c)$ does not exist.

2. For each value of c found in step 1 determine whether $f''(x)$ changes sign at c.

3. If f changes sign at c, the point $(c, f(c))$ is inflection point of f.

Example. Find the intervals of concavity and the inflection point (s)

a) $f(x) = 2x^3 - 3x^2 - 12x$, *b)* $g(x) = x^4 - 6x^2 + 8$ *c)* $g(x) = x\sqrt{x^2 - 4}$

d) $f(x) = \frac{x^2}{x^2+2}$, *e)* $f(x) = (x^2 - 1)^3$ *f)* $f(x) = \ln(1 - \ln x)$ *g)* $f(x) = \frac{e^x}{e^x+2}$ *h)* $f(x) = x^2 \ln x$

Solution: a) $f'(x) = 6x^2 - 6x - 12$, $f''(x) = 12x - 6$ then $f''(x) = 0 \Rightarrow x = 1/2$.

Intervals	$(-\infty, 1/2)$	$1/2$	$(1/2, \infty)$
Sign of $f''(x)$	$-----$	0	$+++++$
concavity of $f(x)$	$f(x)$ is Concave downward		$f(x)$ is Concave upward

Therefore, the graph of f is is Concave downward on $(-\infty, 1/2)$ and Concave upward on $(1/2, \infty)$ and since f'' is change sign at $x = 1/2$ the point $(1/2, f(1/2))$ is inflection point.

Indeterminate Forms and L'Hospital's Rule

L'Hospital's Rule: Suppose f and g are differentiable functions and $g'(x) \neq 0$ near a (except possibly at). Suppose that

$\lim_{x\to a} \boldsymbol{f(x)} = 0$ *and* $\lim_{x\to 0} \boldsymbol{g(x)} = 0$ Or $\lim_{x\to a} \boldsymbol{f(x)} = \pm\infty$ *and* $\lim_{x\to a} \boldsymbol{g(x)} = \pm\infty$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$. Then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (∞ or $-\infty$).

Applying L'Hôpital's Rule

Step 1. Check that the limit of $f(x)/g(x)$ is an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Step 2. Differentiate f and g separately.

Step 3. Find the limit of $f(x)/g(x)$. If this limit is finite, $+\infty$ or $-\infty$.then it is equal to the limit of $f(x)/g(x)$

Example: Evaluate the following limits a) $\lim_{x\to 0}\left(\frac{\ln(x+1)}{x}\right)$ b) $\lim_{x\to 0}\left(\frac{\sin(x)}{x}\right)$

c) $\lim_{x\to 0^+}\left(\frac{\ln(x)}{\frac{1}{x}}\right)$ d) $\lim_{x\to 0}\left(\frac{e^x - x - 1}{x}\right)$ e) $\lim_{x\to 1}\left(\frac{\ln(x)}{x-1}\right)$ f) $\lim_{x\to\infty}\left(\frac{x^2}{e^x}\right)$

Solution: a) since $\lim_{x\to 0}\left(\frac{\ln(x+1)}{x}\right) = \frac{0}{0}$ the we can apply L' Hospital,s rule, thus $\lim_{x\to 0}\left(\frac{\ln(x+1)}{x}\right) = \lim_{x\to 0}\left(\frac{1}{x+1}\right) = 1$. Hence $\lim_{x\to 0}\left(\frac{\ln(x+1)}{x}\right) = 1$.

Indeterminate Pro ducts:

Let $\lim_{x\to a} \boldsymbol{f}(\boldsymbol{x}) = 0$ *and* $\lim_{x\to 0} \boldsymbol{g}(\boldsymbol{x}) = \pm\infty$ then it isn't clear what the value of $\lim_{x\to a} \boldsymbol{f}(\boldsymbol{x})g(x)$, if any, will be. There is a struggle between f and g . If f wins, the answer will be 0 ; if g wins, the answer will be $\pm\infty$. Or there may be a compromise where the answer is a finite nonzero number. This kind of limit is called **an indeterminate form of type** $0.\infty$. We can deal with it by writing the product fgas a quotient:

$$fg = \frac{f}{\frac{1}{g}} \qquad or \qquad fg = \frac{g}{\frac{1}{f}}$$

This converts the given limit into an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$so that we can use l'Hospital's Rule.

Example: Evaluate a) $\lim_{x\to 0^+} (\boldsymbol{x ln}\ (\boldsymbol{x}))$ b) $\lim_{x\to 0^+} \boldsymbol{sin}\ (\boldsymbol{x})\boldsymbol{ln}\ \boldsymbol{x}$ c) $\lim_{x\to 0^+} \frac{x}{csc\, x}$

Indeterminate Differences ($\infty - \infty$): If $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$, the limit $\lim_{x\to a}[f(x) - g(x)] = \infty - \infty$ is called an indeterminate form of type $\infty - \infty$.

Indeterminate forms of type $\infty - \infty$ can sometimes be evaluated by combining the terms or by using a common denominator, or rationalization, or factoring out a common factor and manipulating the result to produce an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Example: Evaluate the following limits a) $\lim_{x\to\infty} (\boldsymbol{x} - \boldsymbol{ln}\ (\boldsymbol{x}))$ b) $\lim_{x\to 0^+} \left(\frac{\boldsymbol{1}}{\boldsymbol{x}} - \boldsymbol{csc}\ (\boldsymbol{x})\right)$

Ans. a) ∞ b) 0

In determinant form of 0^0, 1^∞, $\infty^0$and 0^∞.

Indeterminate forms of types 0^0, 1^∞, $\infty^0$and 0^∞. can sometimes be evaluated by first introducing a dependent variable

$$y = f(x)^{g(x)}$$

and then computing the limit of $ln\, y$. Since $ln\, y = ln[f(x)^{g(x)}] = g(x) \cdot ln[f(x)]$ the limit of $ln\, y$ will be an indeterminate form of type $0 \cdot \infty$ (verify), which can be evaluated by methods we have already studied. Once the limit of ln y is known, it is a straightforward matter to determine the limit of $y = f(x)^{g(x)}$. Or we can follow the rule that $f(x)^{g(x)} = e^{ln(f(x)^{g(x)})} = e^{g(x)ln(f(x))}$ then we can take limit both sides

Example Evaluate a) $\lim_{x\to 0}(\boldsymbol{x^x})$ b) $\lim_{x\to\infty}\left(\mathbf{1}+\frac{\mathbf{1}}{\boldsymbol{x}}\right)^{\boldsymbol{x}}$ c) $\lim_{x\to\infty}(\boldsymbol{e^x}+\boldsymbol{x})^{\frac{\mathbf{1}}{\boldsymbol{x}}}$ d) $\lim_{x\to\infty}\left(\frac{\mathbf{2x-3}}{\mathbf{2x+5}}\right)^{\mathbf{2x+1}}$

e) $\lim_{x\to 0}(\boldsymbol{\sin x})^{\frac{\mathbf{1}}{\boldsymbol{x}}}$ **Ans.** a) 1 b) e c) e d) e^{-8} e) 0

Sketching the graph

Procedure for Graphing *y* = *f*(*x*)

1. **Domain:** determine the domain *D* of *f* , that is, the set of values of for which f is defined.
2. **Intercepts:** The *y*-intercept is $f(0)$ and this tells us where the curve intersects the y-axis. To find the x-intercepts, we set $y = 0$. and solve for x.
3. **Symmetry:**

   i) If $f(x) = f(-x)$ for all *x* in *D*, then *f* is an even function and the curve is symmetric about the *y*-axis.

   ii) If $f(-x) = -f(x)$ for all *x* in *D*, then *f* is an odd function and the curve is symmetric about the origin.

   iii) If $f(x + p) = f(x)$ for all *x* in *D*, where *p* is a positive constant, then *f* is called a periodic function and the smallest such number is called the period.
4. **Asymptotes:**

   i) *Horizontal Asymptotes:* if either $\lim_{x\to\infty}\boldsymbol{f}(\boldsymbol{x}) = L$ or $\lim_{x\to-\infty}\boldsymbol{f}(\boldsymbol{x}) = L$ then the line *y* = *L* is a horizontal asymptote of the curve *y* = *f*(*x*).

   ii) *Vertical Asymptotes:* $\lim_{x\to a^+}\boldsymbol{f}(\boldsymbol{x}) = \pm\infty$ *or* $\lim_{x\to a^-}\boldsymbol{f}(\boldsymbol{x}) = \pm\infty$ then the line *x* = *a* is called vertical asymptote of the curve *y* = *f*(*x*).
5. **Intervals of Increase or Decrease:** Use the Increasing/Decreasing Test
6. **Local Maximum and Minimum Values:** Use the first derivative test or the second derivative test.
7. **Concavity and Points of Inflection:** Compute $f''(x)$ and use the Concavity Test.
8. **Sketch the Curve:** Using the information in items $1-7$, draw the graph.

Example: Sketch the graph of $f(x) = \frac{2x^2}{x^2-1}$ by using above information